

0930 labmeeting

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1 previous

binomial - continuous data 이질성 어떻게 해결할 것인가?

1.1 idea

copula - 를 이용하여 layer 간의 관계 자체를 estimate 하면 특정 layer 가 다른 layer 를 masking 하는 현상을 억제할 수 있지 않을까? why?: copula 를 이용하여 layer 간 상관성을 represent 할 수 있는데, 상관성을 represent 하지 않았을 때는 해당 상관성이 개별 parameter 추정에 개입되었을 수 있다. 해당 상관성을 구분함으로써 parameter 추정을 더 안정적으로 하고, 특정 layer 의 효과가 masking 되는 상황을 피할 수 있지 않을까? 라는 아이디어,,,,

2 Model: LSJM with Copula

2.1 Marginal (per-layer) mean structures

For dyad (i, j) , let x_{ij} be optional covariates. Define linear predictors

$$\eta_{1,ij} = \alpha_1 - \beta_1 d_{ij}, \quad \eta_{2,ij} = \alpha_2 - \beta_2 d_{ij}$$

Binary layer (logit margin). The Bernoulli probability is

$$p_{ij} = \text{logit}^{-1}(\eta_{1,ij}) = \frac{1}{1 + \exp(-\eta_{1,ij})}.$$

Continuous layer (lognormal margin). We parameterize

$$Y_{2,ij} \sim \text{Lognormal}(\mu_{\log,ij}, \sigma_{\log}), \quad \mu_{\log,ij} = \eta_{2,ij} - \frac{1}{2}\kappa_2, \quad \sigma_{\log} = \sqrt{\kappa_2},$$

so that $\mathbb{E}[Y_{2,ij}] = \exp(\eta_{2,ij})$ and $\text{Var}(Y_{2,ij}) = (e^{\kappa_2} - 1) \exp(2\eta_{2,ij})$.

2.2 Copula-based residual coupling (same dyad)

For each dyad (i, j) introduce latent standard normals $(Z_{1,ij}, Z_{2,ij})$ with

$$\begin{bmatrix} Z_{1,ij} \\ Z_{2,ij} \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 & \rho_{ij} \\ \rho_{ij} & 1 \end{bmatrix}\right), \quad \rho_{ij} = \tanh(\kappa_0 + \kappa_1 d_{ij}) \in (-1, 1).$$

The marginal models are preserved via monotone transforms:

Binary (logit margin, thresholding): $q_{ij} = \Phi^{-1}(1 - p_{ij}), \quad Y_{1,ij} = \mathbf{1}\{Z_{1,ij} > q_{ij}\}.$

Lognormal (exact margin via PIT): $U_{2,ij} = \Phi(Z_{2,ij}) \sim \text{Unif}(0, 1), \quad Y_{2,ij} = F_{\text{LN}}^{-1}(U_{2,ij} \mid \mu_{\log,ij}, \sigma_{\log}),$

where Φ is the standard normal CDF and F_{LN} is the lognormal CDF.

Key conditional probability. Using the bivariate normal conditioning identity,

$$Z_{1,ij} \mid Z_{2,ij} = z \sim \mathcal{N}(\rho_{ij}z, 1 - \rho_{ij}^2),$$

hence, with $z_{2,ij} = \Phi^{-1}(F_{\text{LN}}(y_{2,ij} \mid \mu_{\log,ij}, \sigma_{\log}))$,

$$\Pr(Y_{1,ij} = 1 \mid y_{2,ij}) = 1 - \Phi\left(\frac{q_{ij} - \rho_{ij} z_{2,ij}}{\sqrt{1 - \rho_{ij}^2}}\right), \quad q_{ij} = \Phi^{-1}(1 - p_{ij}). \quad (1)$$

2.3 Likelihood

Assuming dyads are conditionally independent given (z, Θ) and that the copula only couples the two layers within the same dyad, the joint likelihood factorizes dyadwise as

$$L(\Theta; Y_1, Y_2) = \prod_{i < j} f_{\text{LN}}(y_{2,ij} \mid \mu_{\log,ij}, \sigma_{\log}) \Pr(Y_{1,ij} = y_{1,ij} \mid y_{2,ij}, \eta_{1,ij}, \rho_{ij}),$$

where the conditional Bernoulli probability is given by (1) (or its complement for $y_{1,ij} = 0$). Therefore the dyad log-likelihood is

$$\ell_{ij} = \log f_{\text{LN}}(y_{2,ij} \mid \mu_{\log,ij}, \sigma_{\log}) + \begin{cases} \log\left[1 - \Phi\left(\frac{q_{ij} - \rho_{ij} z_{2,ij}}{\sqrt{1 - \rho_{ij}^2}}\right)\right], & y_{1,ij} = 1, \\ \log\left[\Phi\left(\frac{q_{ij} - \rho_{ij} z_{2,ij}}{\sqrt{1 - \rho_{ij}^2}}\right)\right], & y_{1,ij} = 0. \end{cases}$$

The total log-likelihood is $\log L = \sum_{i < j} \ell_{ij}.$

2.4 Priors (examples)

$$\begin{aligned}
z_i &\sim \mathcal{N}(0, \sigma_z^2 I_d) \quad (\text{fix } \sigma_z^2 = 1 \text{ or impose } \sum_i \|z_i\|^2 = n \text{ for scale identification}), \\
\alpha_\ell &\sim \mathcal{N}(0, \sigma_\alpha^2), \quad \beta_\ell \sim \mathcal{N}^+(m_\beta, \sigma_\beta^2) \quad (\beta_\ell > 0), \quad \gamma_\ell \sim \mathcal{N}(0, \Sigma_\gamma), \\
\kappa_2 \text{ (lognormal variance)} &: \log \kappa_2 \sim \mathcal{N}(m_\kappa, s_\kappa^2), \\
\kappa &= (\kappa_0, \kappa_1)^\top \sim \mathcal{N}(\mu_\kappa, \Sigma_\kappa).
\end{aligned}$$

3 Posterior Inference via MCMC (no augmentation version)

At each iteration:

1. **Precompute distances and predictors.** Compute d_{ij} from z , then $\eta_{1,ij}, \eta_{2,ij}$, and

$$\mu_{\log,ij} = \eta_{2,ij} - \frac{1}{2}\kappa_2, \quad \sigma_{\log} = \sqrt{\kappa_2}, \quad p_{ij} = \text{logit}^{-1}(\eta_{1,ij}), \quad q_{ij} = \Phi^{-1}(1 - p_{ij}).$$

2. **Continuous layer transform ($z_{2,ij}$).** For each observed $y_{2,ij}$, set

$$u_{2,ij} = F_{\text{LN}}(y_{2,ij} \mid \mu_{\log,ij}, \sigma_{\log}), \quad z_{2,ij} = \Phi^{-1}(u_{2,ij}).$$

3. **Copula correlation.** Set $\rho_{ij} = \tanh(\kappa_0 + \kappa_1 d_{ij})$ (optionally soft-clip $|\rho_{ij}| \leq 0.98$ for stability).
4. **Update (α_1, β_1) (binary layer) (MH)** same as before
5. **Update (α_2, β_2) (lognormal layer) (MH)** same as before
6. **Update κ_2 (lognormal variance).** Use RW–MH in $\log \kappa_2$ space with target

$$\sum_{i < j} \log f_{\text{LN}}(y_{2,ij} \mid \mu_{\log,ij}, \sigma_{\log}) + \log \pi(\kappa_2).$$

7. **Update copula parameters $\kappa = (\kappa_0, \kappa_1)$.** Use RW–MH with proposal $\kappa' \sim \mathcal{N}(\kappa, \Sigma_q)$. The (log) acceptance ratio is

$$\Delta = \sum_{i < j} \left\{ \log \Pr(Y_{1,ij} = y_{1,ij} \mid y_{2,ij}, \eta_{1,ij}, \rho'_{ij}) - \log \Pr(Y_{1,ij} = y_{1,ij} \mid y_{2,ij}, \eta_{1,ij}, \rho_{ij}) \right\} + \log \pi(\kappa') - \log \pi(\kappa),$$

with $\rho'_{ij} = \tanh(\kappa'_0 + \kappa'_1 d_{ij})$. (Equivalently, one may use the bivariate normal copula density on $(Z_{1,ij}, Z_{2,ij})$ in a data-augmentation scheme; see Remark below.)

8. **Update latent positions z_i (MH)** same as before

References